



# MATERIAL SAFETY DATA SHEET

BioPura Propylene Glycol

MSDS DATE: 15.02.2022

FORMER DATE: 16.04.2021

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1 Product identifier

**PRODUCT NAME:** BioPura Propylene Glycol  
**CHEMICAL NAME:** Propane-1,2-diol  
**EC NUMBER:** 200-338-0  
**CAS NUMBER:** 57-55-6  
**REACH REGISTRATION NUMBER:** No registration number is given for this substance since it is exempted from Registration

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

**Identified uses:** Industrial & for professional use only

### 1.3 Details of the supplier of the safety data sheet

**SUPPLIER:** UPM Biochemicals GmbH  
**ADDRESS:** Am Haupttor, Building 4614, 06237 Leuna  
**E-MAIL ADDRESS:** msdsBC@upm.com  
**TELEPHONE:** +49 3461 519 5000

### 1.4 Emergency telephone number

**EMERGENCY NUMBER:** 112  
Poison Information Centres (in Germany), open 24 h daily  
Berlin (responsible for Berlin and Brandenburg) +49 030 19240  
Bonn (responsible for North Rhine-Westphalia) +49 0228 19240  
Erfurt (responsible for Mecklenburg-Western Pomerania, Saxony, Saxony-Anhalt and Thuringia) +49 0361 730 730  
Freiburg (responsible for Baden-Württemberg) +49 0761 19240  
Göttingen (responsible for Lower Saxony, Bremen, Hamburg and Schleswig-Holstein) +49 0551 19240  
Homburg (responsible for Saarland) +49 06841 19240  
Mainz (responsible for Rhineland-Palatinate and Hesse) +49 06131 19240  
Munich (responsible for Bavaria) +49 089 1924

See section 16 for the list of emergency telephone numbers in other countries.

## SECTION 2: HAZARD IDENTIFICATION

### 2.1 Classification of the substance or mixture

Not a hazardous substance or mixture according to the CLP Regulation (EC) No. 1272/2008.

### 2.2 Label elements

None

### 2.3 Other hazards

None

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1 Substances

| EC number: | CAS number: | Chemical name of the substance: | Weight % |             |
|------------|-------------|---------------------------------|----------|-------------|
|            |             |                                 | content  | (range)     |
| 200-338-0  | 57-55-6     | Propane-1,2-diol                | ca. 95.5 | 94.0 – 97.0 |

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## SECTION 4: FIRST AID MEASURES

### 4.1 Description of first aid measures

**EMERGENCY OVERVIEW:** Persons with impaired kidney function may be more susceptible to the effects of this substance. Treat symptomatically and supportively. Consult a physician. Show this safety data sheet to the doctor in attendance.

**RESPIRATION:** Remove the affected person to fresh air. If breathing has stopped, administer artificial respiration. Obtain medical assistance immediately.

**SKIN CONTACT:** Flush the contaminated skin with water. Use soap if available.

**EYE CONTACT:** Rinse immediately with plenty of water, also under the eyelids. Continue irrigation for several minutes while moving eyes to extreme positions. Consult a physician.

**INGESTION:** Rinse mouth.

### 4.2 Most important symptoms and effects, both acute and delayed

The most important known symptoms and effects are described in Section 11.

### 4.3 Indication of any immediate medical attention and special treatment needed

No data available

## SECTION 5: FIREFIGHTING MEASURES

### 5.1 Extinguishing media

Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

### 5.2 Special hazards arising from the substance or mixture

No data available

### 5.3 Advice for firefighters

In the case of fire, if applicable, use flame protective clothing and a respiratory device independent from ambient air. During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1 Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid breathing vapours, mist or gas. Ensure adequate ventilation.

For personal protection see section 8.

### 6.2 Environmental precautions

Do not let product enter drains.

### 6.3 Methods and material for containment and cleaning up

Absorb spill with inert material (e.g. vermiculite, sand or earth), then place in suitable container.

### 6.4 Reference to other sections

Treat the absorbed material according to Section 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1 Precautions for safe handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep container tightly closed.

### 7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place.

### 7.3 Specific end use(s)



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See subsection 1.2.

## SECTION 8: EXPOSURE CONTROL/PERSONAL PROTECTION

### 8.1 Control parameters

#### Occupational exposure limits and/or biological limit values

Do not contain any components with workplace control parameters

### 8.2 Exposure controls

#### Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

#### Personal protective equipment

##### Eye/ Face protection

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

##### Skin protection

Wear appropriate protective gloves to prevent skin exposure.

##### Hand protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands.

##### Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

##### Environmental exposure controls

Do not let product enter drains

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1 Information on basic physical and chemical properties

|                                          |                           |
|------------------------------------------|---------------------------|
| Appearance                               |                           |
| - Physical state:                        | Liquid, viscous           |
| - Color:                                 | Colourless                |
| Odor:                                    | Odorless                  |
| Specific density:                        | 1.0360 g cm <sup>-3</sup> |
| Odor Threshold:                          | No data available         |
| pH value:                                | No data available         |
| Melting point/Freezing point:            | -60 °C                    |
| Initial boiling point/boiling range:     | 187 °C                    |
| Flash point:                             | 99 °C                     |
| Evaporation rate:                        | No data available         |
| Flammability (solid, gas) :              | No data available         |
| Vapor pressure:                          | 0.08 mm Hg @ 20 °C        |
| Lower explosion limit                    | 2.6 vol %                 |
| Upper explosion limit                    | 12.6 vol %                |
| Vapor density:                           | 2.63 - (Air = 1.0)        |
| Solubility(ies) :                        | Miscible with water       |
| Partition coefficient: n-octanol/water : | -1.07                     |
| Auto-ignition temperature:               | >400                      |
| Decomposition temperature:               | No data available         |
| Viscosity:                               | 43.4 mPa·s                |
| Oxidising properties:                    | No data available         |

### 9.2 Other information

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No other information available

## SECTION 10: STABILITY AND REACTIVITY

### 10.1 Reactivity

No dangerous reactivity under normal environmental conditions (air pressure 0.9 to 1.1 bar and temperature -20 °C to +40 °C).  
Hygroscopic: absorbs moisture or water from the air.

### 10.2 Chemical stability

Stable under recommended storage conditions.

### 10.3 Possibility of hazardous reactions

No dangerous reactions are expected when properly used.

### 10.4 Conditions to avoid

Excess heat, moist air.

### 10.5 Incompatible materials

Oxidizing agents.

### 10.6 Hazardous decomposition products

Carbon monoxide, carbon dioxide, aldehyde

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1 Information on toxicological effects

#### ACUTE TOXICITY

LD<sub>50</sub> Oral - rat - 20.000 mg/kg  
LD<sub>50</sub> Oral - rat - 22.000 mg/kg  
LD<sub>50</sub> Dermal - rabbit - 20.800 mg/kg

#### SKIN CORROSION/IRRITATION

Human, skin: Mild irritant

#### SERIOUS EYE DAMAGE/IRRITATION

Rabbit, eye: Mild irritant (Draize test)

#### RESPIRATORY OR SKIN SENSITISATION

Not sensitizing

#### GERM CELL MUTAGENICITY

Not mutagenic

#### CARSINOGENICITY

Not identified as probable, possible or confirmed human carcinogen by IARC

#### REPRODUCTIVE TOXICITY

Not a reproductive toxicant

#### STOT-SINGLE EXPOSURE

Not toxic after single exposure

#### STOT-REPEATED EXPOSURE

Not toxic after repeated exposure.

#### ASPIRATION HAZARD

Aspiration hazard is not expected.

#### POTENTIAL HEALTH EFFECTS:

May cause reversible Gastrointestinal disturbance, Nausea, Headache, Vomiting, Central nervous system depression

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## SECTION 12: ECOLOGICAL INFORMATION

### 12.1 Toxicity

Short-term toxicity:

Invertebrates: 48-h EC50 (immobilisation; Daphnia magna): > 10 000 mg/L  
48-h NOEC (mortality; Daphnia magna): > 13 020 mg/L

Fish: 96-h NOEC (mortality; Fathead minnow): 52 930 mg/L

### 12.2 Persistence and degradability

Readily biodegradable

### 12.3 Bioaccumulative potential

Based on the log Pow value, the substance has no bioaccumulation potential.

### 12.4 Mobility in soil

Based on the result of the octanol/water partition coefficient, the substance is expected to have a low mobility in soil.

### 12.5 Results of PBT and vPvB assessment

The substance is not PBT / vPvB.

### 12.6 Other adverse effects

No further information is available.

### 12.7 Additional information

None.

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1 Waste treatment methods

Waste should only be disposed of via a licensed waste contractor.

### Treatment of contaminated packaging

A specific treatment is not required.

### Waste codes

There is no EC waste code for this material according to the guideline 75/422/ECC.

### Specific precautionary measures

None.

### Relevant EC or other regulations

None.

## SECTION 14: TRANSPORT INFORMATION

### 14.1 UN number

Not regulated.

The material is not considered a hazardous good

### 14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

### 14.3 Transport hazard class(es)

None.

### 14.4 Packing group



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None

## 14.5 Environmental hazards

Based on available data, the classification criteria are not met.

## 14.6 Special precautions for user

See section 7.

## 14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

No data available

## SECTION 15: REGULATORY INFORMATION

### 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1907/2006 (REACH)

### 15.2 Chemical safety assessment

In accordance with Regulation (EC) No. 1907/2006 (REACH) Article 14, a Chemical Safety Assessment has been carried out for this substance.

## SECTION 16: OTHER INFORMATION

### 16.1 Additions, Deletions, Revisions

Version 1.0.

This safety data sheet is drawn up to comply with the requirements Annex II of Regulation (EC) No. 1907/2006 (REACH), as amended by Annex I to Commission Regulation (EU) No. 453/2010 of 20 May 2010.

### 16.2 Key or legend to abbreviations and acronyms

- CLP - Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on classification, labelling and packaging of substances and mixtures
- LD50: Lethal dose of the substance that causes 50 % mortality of the test population
- OSHA - Occupational Safety and Health Administration (USA)
- NOEC - No observed effect concentration
- PBT/vPvB - Persistent, bioaccumulative and toxic/ very persistent and very bioaccumulative
- REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
- STOT RE - Specific Target Organ Toxicity, Repeated Exposure
- STOT SE - Specific Target Organ Toxicity, Single Exposure

### 16.3 Emergency telephone number

Europe wide emergency number 112

USA, Canada, Brazil 911

Poison information centres **AUSTRIA** (Vienna, Wien) +43 1 406 43 43; **BELGIUM** (Brussels, Bruxelles) +32 70 245 245; **BULGARIA** (Sofia) +359 2 9154 409; **CZECH REPUBLIC** (Prague, Praha) +420 224 919 293; **DENMARK** (Copenhagen) 82 12 12 12; **ESTONIA** (Tallinn) 112; **FINLAND** (Helsinki) +358 9 471 977; **FRANCE** (Paris) +33 1 40 0548 48; **GERMANY** (Berlin) +49 30 19240; **GREECE** (Athens, Athina) +30 10 779 3777; **HUNGARY** (Budapest) 06 80 20 11 99; **ICELAND** (Reykjavik) +354 525 111 or +354 543 2222; **IRELAND** (Dublin) +353 1 8379964; **ITALY** (Rome) +3906 305 4343; **LATVIA** (Riga) +371 704 2468; **LITHUANIA** (Vilnius) +370 5 236 20 52 or +370 687 53378; **MALTA** (Valletta) 2425 0000; **NETHERLANDS** (Bilthoven) +31 30 274 88 88; **NORWAY** (Oslo) 22 591300; **POLAND** (Gdansk) +48 58301 65 16 or +48 58 349 2831; **PORTUGAL** (Lisbon, Lisboa) 808 250 143; **ROMANIA** (Bucharest) +40 21 3183606; **SLOVAKIA** (Bratislava) +421 2 54 77 4166; **SLOVENIA** (Ljubljana) + 386 41 650500; **SPAIN** (Barcelona) +34 93 227 98 33 or +34 93 227 54 00 bleep 190; **SWEDEN** (Stockholm) 112 or +46 833 12 31 (mon-fri 9.00-17.00); **UNITED KINGDOM** (London) 112 or 0845 4647 (NHS Direct).

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